PPS 12.1

1 B

2 D

3 D

4 C

5 C

6a) Under grating:

$\lambda = d\sin 30^\circ$
$d = 2\lambda$

Under double slit:

$\lambda = 500 d \sin\theta$

$\therefore 500 d \sin\theta = \frac{d}{2}$

$\sin\theta = \frac{1}{1000}$

$\theta = 0.0573$

$\rightarrow \tan\theta = \frac{W}{D} = W \quad (D=1)$

$W = \tan(0.0573)$
$= 10^{-3}$

$\therefore W = 1.0\,mm$

b) $\theta_g = 30^\circ$

$\therefore W_g = 1 \times \tan 30^\circ = \frac{1}{\sqrt{3}} = 0.5774\,m$

$\theta_s = 0.001^\circ$

$\rightarrow W_g = D \times \tan 0.001^\circ$

$D = \frac{0.5774}{10^{-3}}$

$= 580\,m$

7a) Wavefronts will shift as H moves closer to R   let this wave be W_R
Wavefronts stay fixed when wave reflects against M   let this one be W_M

As W_R decreases, $\Delta\lambda$ varies between 0 and π
As $I \propto A^2$, I varies between I_{min} and I_{max} every $\frac{1}{2}\lambda$ (as $\Delta\lambda$ varies between 0 and π)
So intensity varies as λ varies $\rightarrow$ As H moves closer to R, λ varies so I varies.

7 b) λ: if λ is too small, (or too large) then distance between I_{max} and I_{min} will be too small (or too large) to reasonably measure (i.e. f is too low (1mm) or too high ($\sim 10^{7}$ m) as v is constant)

if the distance between T and R is too large, as H moves away from M the distance each interference max/min moves a larger distance away the further T and R are apart.

If the starting distance that T and R are from M is too small, the ΔPath between the waves reflected off M and the waves reflected off H is will be very small, and H will have to vary by a very small amount, increasing % error.

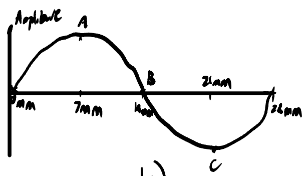
c) $\lambda = 28$mm

$f = \frac{v}{\lambda}$

$= \frac{3\times10^{8}}{28\times10^{-3}}$

$= 10.7$ GHz

7mm between maxima and minima is correct:



At A, I (intensity) is at its maximum ($I \propto A^2$)
At B, I is at its minimum, B − A = 7mm
so ΔPath between I_{max} and I_{min} = 7mm.

At C, I is maximum because $I = k(A)^2$
$= kA^2$
$\therefore I \propto A^2$ $(A^2 > 0)$

As H moves towards T and R at 2.8 ms^{-1}:

$2.8 ms^{-1} = 2800\ mms^{-1}$

Intensity oscillates between max & min every 7mm

↳ every 2.5 ms (at 2.8 ms^{-1})

$f = \frac{1}{T}$

$T = 2.5 ms^{-1}$

$\therefore f = \frac{1}{2.5\times10^{-3}} = 400$ Hz.

So sound will be at 400 Hz, not 200 Hz.

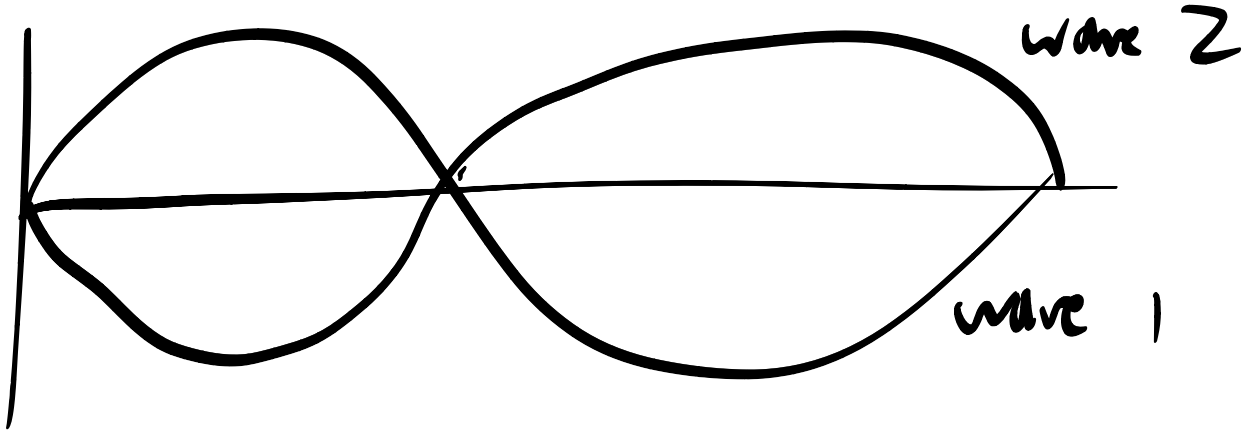
8a) interference occurs when two coherent waves are superposed.

b) Δ Path $= 15 - 10.5 = 4.5$ cm
$\lambda = 3$ cm

$\Delta \text{Path} = \frac{3}{2}\lambda$

$\therefore$ negative interference

Waves are out of phase by 180° or π



So at every point on the wave both waves' amplitudes sum to 0
$\therefore$ negative interference

9 10 peaks measures 5.4cm across
peak - peak is 0.6cm

$n\lambda = d\sin\theta$
$\frac{o}{a} = \tan\theta$

$\theta = \arctan\left(\frac{5.4\times10^{-2}}{4.25}\right)$
$= 0.72795°$

$n\lambda = 0.25\times10^{-3} \times \sin(0.72795)$
$9\lambda = 3.18\times10^{-6}$
$\lambda = 353$ nm

red light has lower wavelength
$\therefore \lambda$ increases
d is constant $\therefore \theta$ increases

so fringes get further apart

10 a) 300 lines /mm
$d = \frac{1}{300}$ mm / line

$D = 5.0$ m

To red : $700 \times 10^{-9} = \frac{1}{300} \sin\theta_r$
$\sin\theta_r = 210 \times 10^{-6}$

To purple $400 \times 10^{-9} = \frac{1}{300} \sin\theta_p$
$\sin\theta_p = 120 \times 10^{-6}$

width $= 5\sin\theta_r - 5\sin\theta_p$

$= 450 \times 10^{-6}$
$= 0.45$ mm width

b) second order rainbow overlaps with third order

Mathematically:

second order red fringe is further away from center than third order purple.

red: $5 \times 2 \times 210 \times 10^{-6} = 2100 \times 10^{-6} = 2.1 \times 10^{-3}$

purple: $5 \times 3 \times 120 \times 10^{-6} = 1800 \times 10^{-6} = 1.8 \times 10^{-3}$

$\therefore$ second order